Signals and Systems — Tutorial 8

Filtering

Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a $\star$.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

Prework questions

A noisy signal, $f(t)$, shown in Figure 1, filtered using the filters $F_1(s), F_2(s)$ and $F_3(s)$ with Bode diagrams as shown in Figure 2. The output of each filter is shown in Figure 3 by $y_g(t), y_b(t)$ and $y_r(t)$ given by the green, blue and red lines respectively.

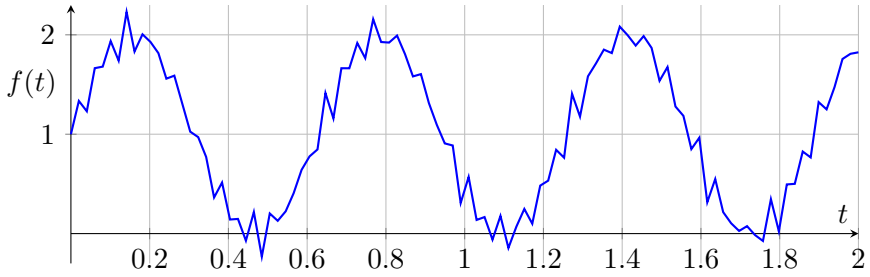


Figure 1: Measured signal, $f(t)$

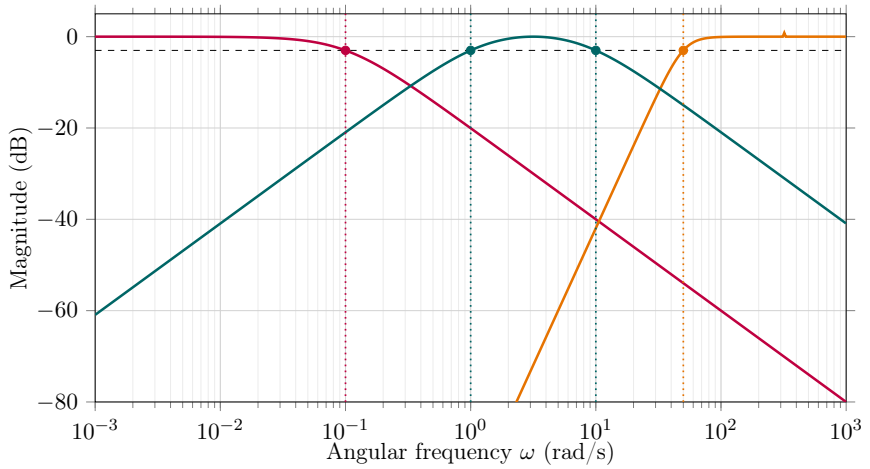


Figure 2: Bode plots of the filters $F_1(s), F_2(s)$ and $F_3(s)$.

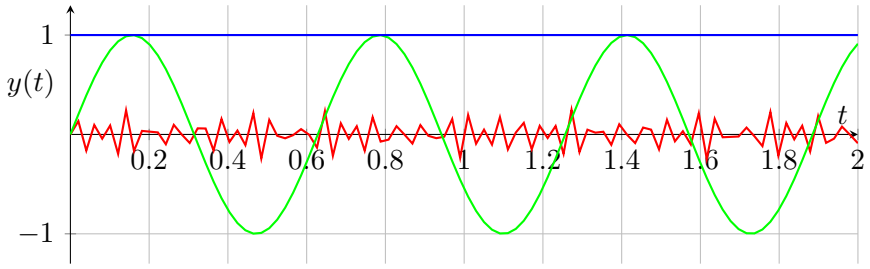


Figure 3: Filtered outputs $y_g(t)$ (green), $y_b(t)$ (blue) and $y_r(t)$ (red) of the measured signal $f(t)$.

Identify whether F_1, F_2 and F_3 are low-pass, high-pass, band-pass or band notch filters and match the filtered outputs $y_g(t), y_b(t)$ and $y_r(t)$, shown in Figure 3 to which filter was used to filter the signal, $f(t)$.

Tutorial questions

1: Poles and zeros

1. For each of the filters F_1, F_2 and F_3 in Figure 2 identify the location of the poles and zeros of the filter and hence find the transfer functions for each of the filters.
2. Find the time domain differential equation for each of the filters and determine their impulse responses.
3. For each of the filters find the frequency ω_c such that a sinusoidal input $A\sin(\omega_c t)$ is attenuated by a factor of $1/\sqrt{2}$. Demonstrate this by using the convolution for at least one of the filters.

2: RLC Filter

Consider the RLC series circuit in Figure 4 with input $v(t)$. Determine if the circuit is a high-pass, low-pass or bandpass filter if the output is **(a)** the voltage across the inductor V_L, **(b)** the voltage across the capacitor V_C or **(c)** the current i in the circuit and determine the cut-off frequencies for each case.

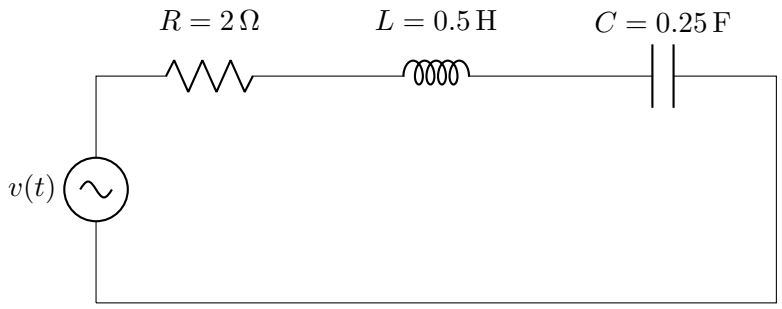


Figure 4: RLC series circuit.

3: Filters in Cascade

Consider the circuit (two first order filter stages connected in cascade) in Figure 5.

1. Using the transfer functions

$$G_A(s) = \frac{V_A(s)}{U_A(s)} \text{ and } G_B(s) = \frac{V_B(s)}{U_B(s)}$$

   Show that stage A is a high-pass filter, stage B is a low-pass filter and determine thier DC gains.

2. Determine values of R, L and C so that the circuit has the frequency response as shown by the green curve in Figure 2.

4: Butterworth filters

Suppose the orange curve in Figure 2 is the frequency response of an order n Butterworth filter.

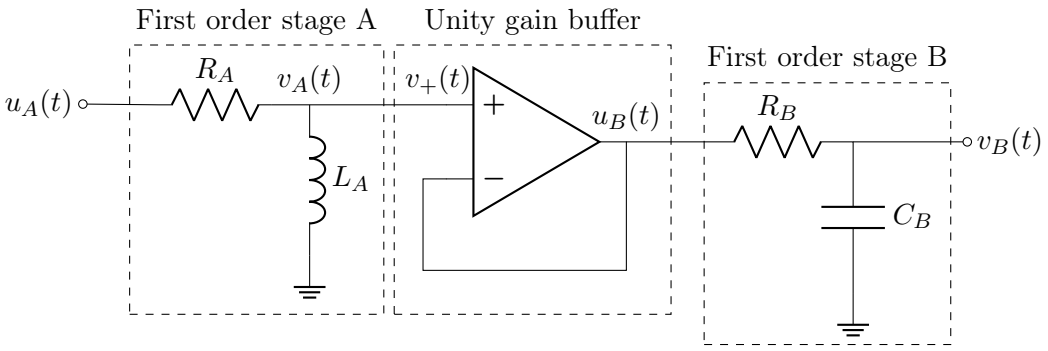


Figure 5: Two first order filter stages (implemented as RL and RC circuits) connected in cascade by a unity-gain voltage buffer i.e. a second order band-pass/stop filter

1. Is the filter a lowpass, highpass, bandpass or bandstop filter?
2. Determine the order n of the filter.
3. Find the damping coefficients ζ_k and give the transfer function of the filter.
4. Find the phase of the filter at the cut-off frequency, ω_c.

5: Filter Design - Tunable Suspension

Mechanical engineers have designed a tunable suspension system for a car based on a mass-spring-damper where the magnitude of the damping coefficient is adjustable. The suspension system satisfies

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = F(t) \tag{1}$$

where m is the mass of the car, b is the damping coefficient, k is the spring constant, $x(t)$ is the vertical displacement of the car and $F(t)$ is some external force.

1. Show that (1) is a low-pass filter and determine its DC gain and cut-off frequency in terms of m, b and k.
2. The car is travelling at 36 km/h over a very rough road with potholes every 10 cm. In order not to damage the car the effect of this needs to be attenuated by 100 dB. However, speed-bump regulations specify that the external force from speed-bumps placed at one metre intervals can be attenuated by no more than 20 dB. Determine the minimum order of a filter which meets these specifications and give its transfer function.
3. To build this filter specify how many mass-spring-damper systems must be put in cascade with damping coefficients b_i and spring constants k_i and give values for each which realises the filter from above assuming the mass of the car is 1000 kg.
4. The mechanical engineers want to use identical springs and dampers for each of the mass-spring-damper systems used. Is this possible? Determine if the mechanical engineers should have paid more attention in ELEC2302.

6: Smoothing Filter

Recall from Tutorial 5 that a DC power supply takes a 50 Hz mains sinusoid, passes it through the full-wave rectifier of Tutorial 3 and then through the LC filter of Figure 6 to remove the ripple where $R \approx 0.5\,\Omega$ is the winding resistance of the inductor.

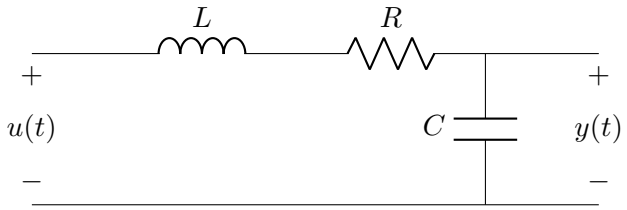


Figure 6: Smoothing filter. The input $u(t)$ is the rectifier output and $y(t)$ is the supply output.

The rectifier output is $u(t) = A\,|\sin(\omega_m t)|$ with $\omega_m = 2\pi \times 50$ rad/s. A measure of the ripple is the RMS of the non-constant parts of the signal i.e.

$$RMS(f) = \sqrt{\frac{1}{T}\int_0^T |f(t) - c_0|^2\,dt} \tag{2}$$

Recall the Fourier series of a rectified sine wave is

$$A|\sin(\omega_0 t)| = \sum_k c_k e^{j\omega_0 k t} = \sum_k \frac{e^{j\omega_0 k t}}{\pi\,(1-4k^2)}$$

1. Find the magnitude of the ripple of the unfiltered rectified wave.

2. Using the Fourier series of the rectified sine from Tutorial 3, write down the DC term and the amplitude of the lowest-frequency ripple component, and state the frequency of that component. Show that the ratio of ripple amplitude to DC level at the filter input is 2/3, independent of A.

3. The Fourier transform of a periodic signal is a train of impulses,

$$U(j\omega) = 2\pi \sum_k c_k\,\delta\,(\omega - k\omega_0)\,,$$

   where ω_0 is the fundamental and c_k are the complex Fourier coefficients. Write down the c_k of $u(t)$ when $A = 1$ and hence the weight of each impulse. Plot the magnitude and phase of $U(j\omega)$ for $-6\omega_m \le \omega \le 6\omega_m$. The ratio of the first ripple line to the DC line is not the 2/3 of Part 1. Explain.

4. Find $G(j\omega) = Y(j\omega)/U(j\omega)$. Put $G(j\omega)$ in the standard second-order form and identify ω_n, ζ and $G(0)$.

5. Sketch $|G(j\omega)|$ in decibels against $\log_{10}\omega$. Mark on the same axes the frequency of the ripple component found in Part 1.

6. The supply must deliver a DC level with ripple no greater than 1% of that level. Show that this requires $|G(j2\omega_m)| \le 0.015$. Using the Bode diagram from part 5, find the largest ω_n that meets the specification. If $L = 10$ mH, choose a value of C that achieves the specification.

7. Using $Y(j\omega) = G(j\omega)U(j\omega)$ and the sifting property, write down the weight of each line of $Y(j\omega)$. Evaluate the DC line and the first three ripple lines and plot the magnitude of $Y(j\omega)$ on the axes of part 3.